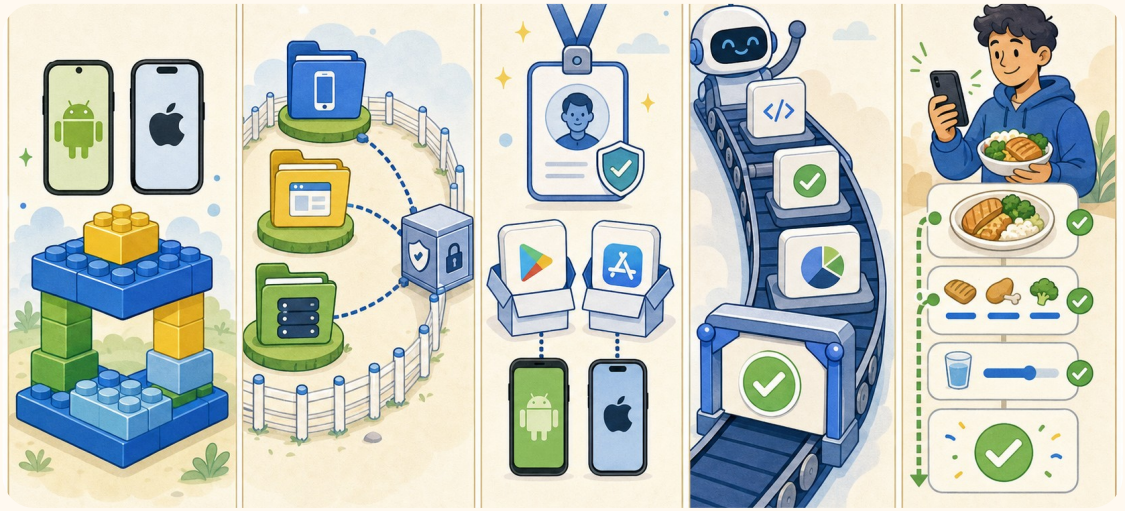
SUPERNUTRITION

AI nutritionist

22 stories.
One clear journey.

From a trusted mobile foundation to a photo-based nutrition result.

6 epics · 22 stories · 1 flow







E1

Foundation & product shell

Make the repo safe to build on



5 stories

E1.1 E1.2 E1.3 E1.4 E1.5

E2

Capture & photo preparation

Get one safe photo into the flow



3 stories

E2.1 E2.2 E2.3

E3

Shared inference & local engines

Keep analysis local when possible



4 stories

E3.1 E3.2 E3.3 E3.4

E4

Remote inference & consent

Make upload explicit and bounded



3 stories

E4.1 E4.2 E4.3

E5

Confirmation & grounded nutrition

Turn evidence into a trusted result



3 stories

E5.1 E5.2 E5.3

E6

Journey evidence & evaluation

Prove the flow on every platform



4 stories

E6.1 E6.2 E6.3 E6.4

E1

E1.1

Activate the mobile KMP scaffold



SHIPS

Pinned Android ✦ iOS roots

DONE WHEN

Thin hosts ✦ bootstrap proof

FLOW

Pin tuple → Prove hosts

E1

E1.2

Enforce monorepo boundaries and contracts



SHIPS

Fenced product roots

DONE WHEN

Contract matrix ✦ policy inventory

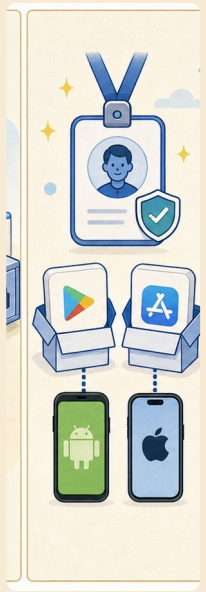
FLOW

Fence roots — Test contracts

E1

E1.3

Establish product identity metadata



SHIPS

One identity manifest

DONE WHEN

SuperNutrition labels stay aligned

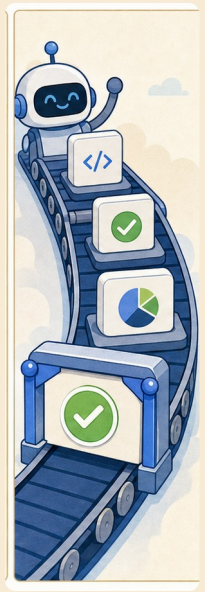
FLOW

Set identity — Match stores

E1

E1.4

Establish repository quality gates



SHIPS

Read-only checks ✦ push CI

DONE WHEN

85% coverage; no green skips

FLOW

Run checks — Block gaps

E1

E1.5

Establish the low-friction interaction policy



SHIPS

Defaults ✦ interaction budget

DONE WHEN

Shortest safe path is encoded

FLOW

Prefill — Keep control

E2

E2.1

Implement camera and picker bridges



SHIPS

Capture or import one photo

DONE WHEN

Permission recovery has a way out

FLOW

Open camera — Recover

E2

E2.2

Normalize and quality-check Photo Input



SHIPS

Safe, equivalent photo bytes

DONE WHEN

Orientation ✦ metadata are clean

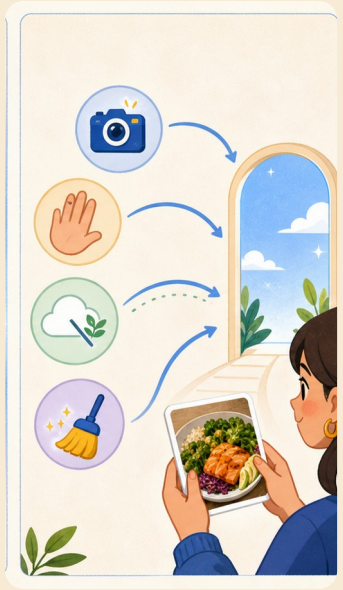
FLOW

Normalize — Scrub

E2

E2.3

Implement capture recovery and session cleanup



SHIPS

Retake / cancel / offline recovery

DONE WHEN

No stale result or retained bytes

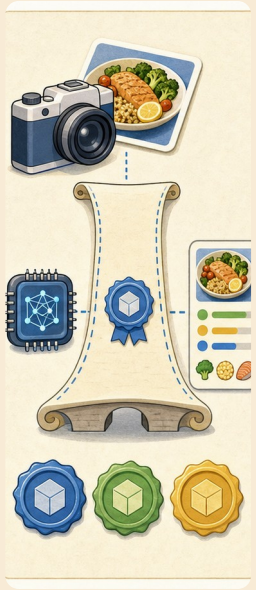
FLOW

Recover — Clean

E3

E3.1

Freeze contract-v1 inference ports



SHIPS

Shared request ✦ candidate types

DONE WHEN

Fixtures; no nutrition in engines

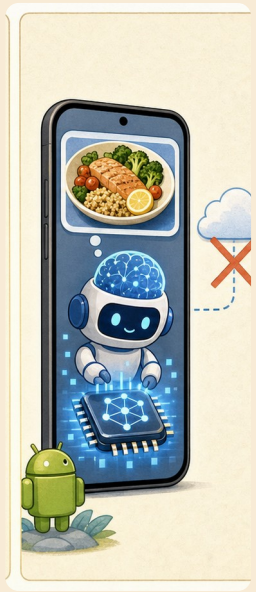
FLOW

Define — Freeze

E3

E3.2

Implement local Android inference



SHIPS

On-device Android analysis

DONE WHEN

Zero photo network traffic

FLOW

Load model → Infer local

E3

E3.3

Implement local iOS inference



SHIPS

On-device iOS analysis

DONE WHEN

Same evidence vocabulary

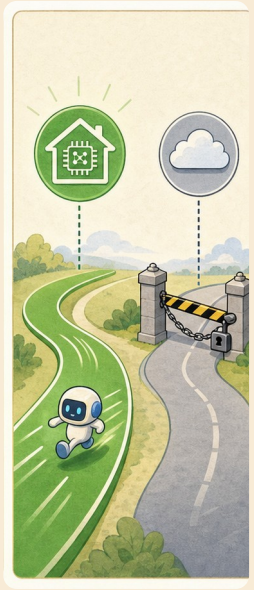
FLOW

Load model — Infer local

E3

E3.4

Integrate deterministic local-first routing



SHIPS

Bounded route policy

DONE WHEN

Remote only when it can help

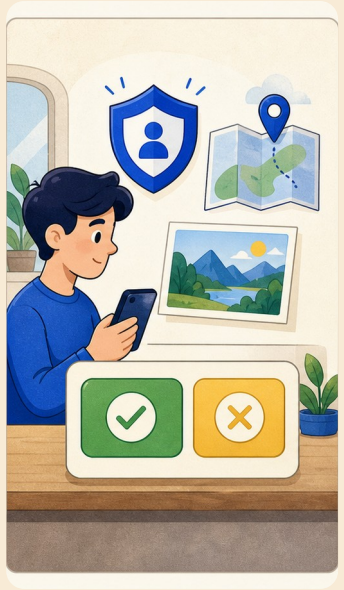
FLOW

Try local — Escalate

E4

E4.1

Implement remote consent and transfer states



SHIPS

Clear upload choice

DONE WHEN

Pre-transfer cancel sends zero bytes

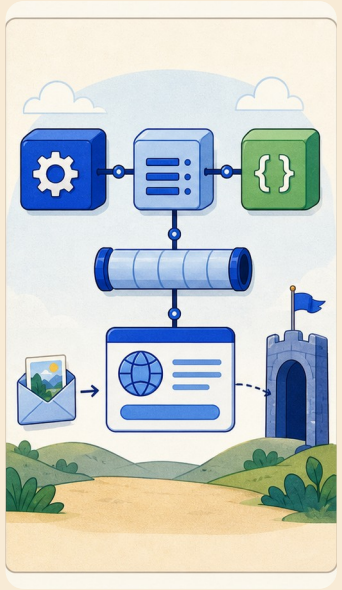
FLOW

Explain — Consent

E4

E4.2

Implement the remote mobile adapter



SHIPS

Ktorfit API over Ktor

DONE WHEN

Bounded, redacted, idempotent transfer

FLOW

Generate API — Upload safely

E4

E4.3

Implement the first-party gateway boundary



SHIPS

Provider isolation ✦ policy gate

DONE WHEN

No durable raw photos

FLOW

Validate — Dispose

E5

E5.1

Implement Candidate review and correction



SHIPS

Inspect / remove / replace / search

DONE WHEN

User remains the authority

FLOW

Review → Confirm

E5

E5.2

Implement country-aware nutrition catalog selection



SHIPS

Catalog ≠ processing region

DONE WHEN

Change invalidates mapped result

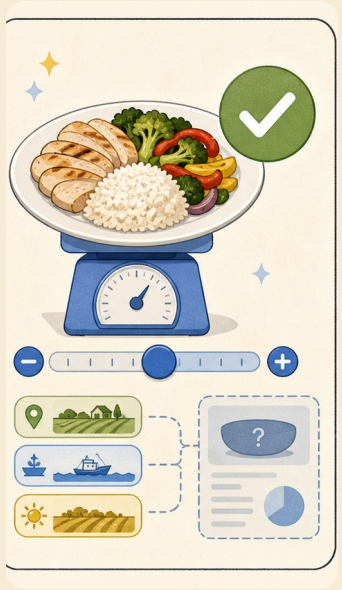
FLOW

Choose catalog — Re-map

E5

E5.3

Implement serving confirmation and grounded result



SHIPS

Portion ✦ assumptions

DONE WHEN

No ungrounded nutrition numbers

FLOW

Set serving — Ground result

E6

E6.1

Build the shared UI journey DSL



SHIPS

One logical flow; two platforms

DONE WHEN

Same journey definitions run twice

FLOW

Define once — Run twice

E6

E6.2

Add privacy/network and native-gap oracles



SHIPS

Binary-bound evidence

DONE WHEN

UI green cannot hide native failure

FLOW

Observe → Bind proof

E6

E6.3

Establish the evaluation corpus and route baselines



SHIPS

Local / remote / hybrid baselines

DONE WHEN

G-5 thresholds before confidence routing

FLOW

Measure → Set gates

E6

E6.4

Maintain the quality and release evidence gates



SHIPS

Freshness-aware CI gate

DONE WHEN

Stale or missing evidence blocks release

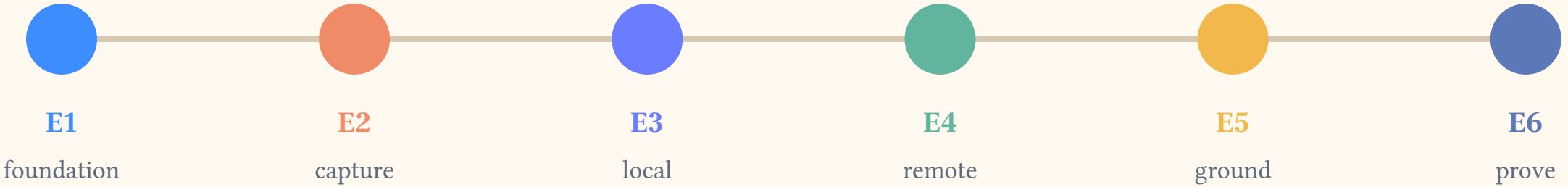
FLOW

Check evidence → Release

The finish line is a trusted meal scan

Capture → understand → confirm → ground

Every epic removes one kind of uncertainty. Together they make the simplest safe path the default.



SuperNutrition: AI nutritionist